Bin Hu

Networked Systems & Machine Learning | Ph.D. Student, USC
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Hbbbba.github.io/Resume

EDUCATION

University of Southern California (USC)

Ph.D. in Electrical and Computer Engineering | Aug. 2023 - May 2028 (Expected)

Advisor: Prof. Feng Qian. Expected graduation: 05/2028.

Prior doctoral studies: University of Minnesota, CSE, Aug. 2022-Aug. 2023; transferred with advisor.

Southeast University (SEU)

M.Sc. in Computer Science and Engineering | Sept. 2019 - Jun. 2022

Southeast University (SEU)

B.Eng. in Computer Science and Engineering | Sept. 2015 - Jun. 2019

TECHNICAL SKILLS

Programming: Python, C/C++, Java, MATLAB.

Tools & platforms: PyTorch, Linux, WebRTC, Android network measurement, packet capture/analysis, GNURadio, SDR.

RESEARCH & ENGINEERING EXPERIENCE

University of Southern California / University of Minnesota

Ph.D. Student | Aug. 2022 - Present

- Contribute to Net-JEPA, a 4M-parameter self-supervised network world model for adaptive bitrate streaming and congestion control, combining latent prediction, linear readouts, and model predictive control.
- Framework evaluation: pretraining on 450,000+ Puffer streams; frozen-model transfer to congestion control with 20 calibration traces achieved capacity forecast MSLE of 0.84 vs. 1.93 for random initialization in fluid-queue emulation.
- Co-authored a satellite-link prediction framework combining geometry and telemetry (HotMobile 2026). Real-Starlink evaluation reported throughput/latency MAE of 8.37 Mbps/14.25 ms vs. 29.54 Mbps/26.31 ms for a self-similar baseline.
- Co-authored prediction-driven video transmission and reconstruction over LEO links (INFOCOM 2026 Poster); developed WebRTC video-quality measurement, Raspberry Pi multipath relays, and an Android latency/packet-loss measurement app.

SUTD | Research Assistant | May 2021-Apr. 2022; Jul. 2021

- Developed interpretable ML models for modulation classification and implemented an AI-based software modulator.

Southeast University | Graduate Researcher | Sept. 2019-Oct. 2020; Jun. 2021

- Designed and evaluated a multidimensional LoRa demodulator and implemented a fine-grained self-correlation collision detector (ICNP 2020; WASA 2021).

SELECTED PUBLICATIONS

Predicting Connectivity and Link Performance in Dynamic LEO Satellite Networks. **ACM HotMobile, 2026.** Bin Hu et al.

Prediction-Driven QoE Optimization for Video Calls over LEO Satellite Networks. **IEEE INFOCOM Poster, 2026.** Bin Hu et al.

NN-Defined Modulator: Reconfigurable and Portable Software Modulator on IoT Gateways. **USENIX NSDI, 2024.** Jiazhao Wang et al. (co-author: Bin Hu)

SCLoRa: Leveraging Multi-Dimensionality in Decoding Collided LoRa Transmissions. **IEEE ICNP, 2020.** Bin Hu et al.

SELECTED HONORS

Student Travel Grant, ACM HotMobile 2026